Lesson 1: Absolute Value of Real Numbers

Assignment

1. Evaluate.

(a) $|-23|$ (b) $-|17|$ (c) $-|-0.6|$ (d) $\left|-\frac{5}{9}\right|$

(e) $\sqrt{(-12)^2}$ (f) $-\sqrt{8^2}$ (g) $|-\pi|$ (h) $-|-(-3)|$

2. Evaluate.

(a) $|7-19|$ (b) $|7|-|19|$ (c) $|-8-(-3)|$ (d) $4|-2|-|-10|$

(e) $-3|6-11|+5$ (f) $\big||-4|-|9|\big|$ (g) $|-2|^3-|-2^3|$ (h) $\dfrac{|-6|+2}{|-4|}$

3. Evaluate. Give exact values, and also the value to the nearest hundredth in (e) and (f).

(a) $|\sqrt[3]{-1000}|$ (b) $-|-\sqrt[3]{343}|$ (c) $|\sqrt{25}-\sqrt{144}|$

(d) $|\sqrt[3]{-8}-\sqrt{9}|$ (e) $|3-\sqrt{11}|$ (f) $|\sqrt{5}-\sqrt{7}|$

4. Evaluate each expression for $a=-5$ and $b=2$.

(a) $|a-b|$ (b) $|a|-|b|$ (c) $|3ab|$

(d) $|a^2-4b|$ (e) $|2a+3b|$ (f) $\dfrac{|a+b|}{a}$

5. Arrange in order from least to greatest. $|-4.2|, \quad -|-5|, \quad |1-6|, \quad |-\sqrt{17}|, \quad -|0.8|, \quad \left|-\frac{13}{3}\right|$

6. Replace each $\square$ with $<$, $>$ or $=$.

(a) $|-9| \square |9|$ (b) $|-3-5| \square |-3|+|5|$ (c) $|4-10| \square |4|-|10|$

(d) $|-2+7| \square |-2|+|7|$ (e) $-|-6| \square |-6|$ (f) $|(-3)(-4)| \square |-3| \cdot |-4|$

7. True or false? If true, explain; if false, give a counterexample.

(a) The absolute value of every real number is positive.

(b) $|a+b| = |a|+|b|$ for all real numbers a and b.

(c) $-|x|$ is never positive.

(d) If $|a| = |b|$, then $a = b$.

8. Determine the distance between the numbers. Give exact answers.

(a) 13 and 4 (b) -7 and 8 (c) -15 and -6

(d) -3.4 and 2.9 (e) $-\frac{1}{2}$ and $\frac{5}{3}$ (f) $\sqrt{3}$ and $-2\sqrt{3}$

9. Use the idea of distance on a number line.

(a) Point A is at -2. Point B is 7 units from A. State the possible positions of B.

(b) $|x-3|$ is the distance between x and 3. Determine all values of x for which $|x-3| = 5$.

(c) Determine all values of x for which $|x+6| = 1$.

10. A stock's closing price changed by +\$1.25, −\$0.80, −\$2.10, +\$0.45 and +\$1.60 over five days.

(a) Determine the net change in the price over the five days.

(b) Determine the total amount the price moved (up or down) over the five days.

(c) The price was \$38.75 before the first day. What was it after the fifth day?

11. A cereal box is labelled 750 g. A box with mass m grams passes inspection if $|m - 750| \leq 12$.

(a) Which of these boxes pass: 739 g, 762 g, 763.5 g, 737 g?

(b) State the lightest and the heaviest mass that passes.

12. The piecewise definition of absolute value.

(a) Complete: $|x| = \begin{cases} _____, & \text{if } x \geq 0 \\ _____, & \text{if } x < 0 \end{cases}$

(b) True or false? (i) $|x| = -x$ if $x < 0$ (ii) $|x| = x$ if $x \leq 0$ (iii) $|-x| = x$ if $x \geq 0$ (iv) $|-x| = -x$ if $x > 0$

(c) Explain why the statement in (b)(i) does not give a negative absolute value.

13. Simplify each expression for the values of x given.

(a) $5x - |x|,\ x < 0$ (b) $|x| + |-x|,\ x > 0$ (c) $\dfrac{|3x|}{x},\ x < 0$ (d) $\sqrt{x^2} + x,\ x < 0$

14. Write $|x + 5|$ in piecewise form. Explain what $|x + 5|$ measures on a number line, and evaluate it for $x = -8$ and for $x = 1.5$ using your piecewise form.

15. The value of $-|3 - 8| + |-2|$ is

A. 7 **B.** -3 **C.** 3 **D.** -7

16. Which statement is true for every real number x?

A. $|x| = x$ **B.** $|-x| = x$ **C.** $\sqrt{x^2} = |x|$ **D.** $|x| > 0$

17. The numbers on a number line that are 6 units from -2 are

A. 4 only **B.** -8 and 4 **C.** -4 and 8 **D.** -6 and 6

18. Numerical Response The distance between -2.65 and 4.9 on a number line is _____.

Lesson 2: Graphing Absolute Value Functions of Linear Functions

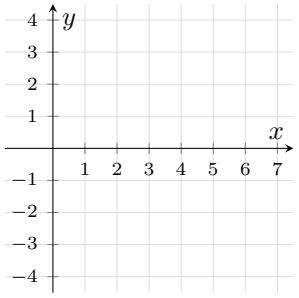
Assignment

1. Let $f(x) = 3 - x$.

(a) Complete the table, then graph $y = f(x)$ and $y = |f(x)|$ on the grid.

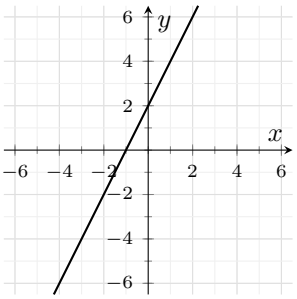
(b) State the invariant point and describe which part of the line was reflected.

x	0	1	2	3	4	5	6
$y = f(x)$							
$y = \|f(x)\|$							

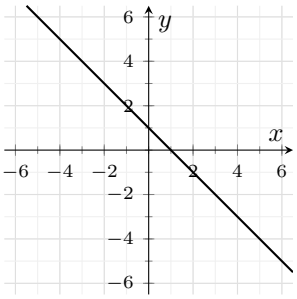


2. The graph of $y = f(x)$ is shown. Sketch the graph of $y = |f(x)|$ on the same grid and state the invariant point(s), if any.

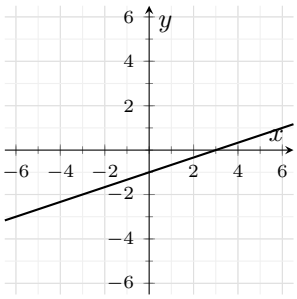
(a) $f(x) = 2x + 2$



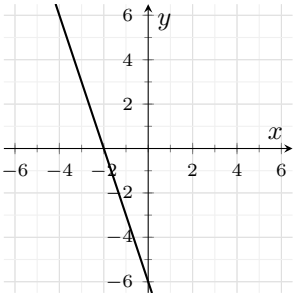
(b) $f(x) = -x + 1$



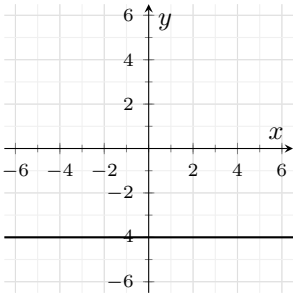
(c) $f(x) = \frac{1}{3}x - 1$



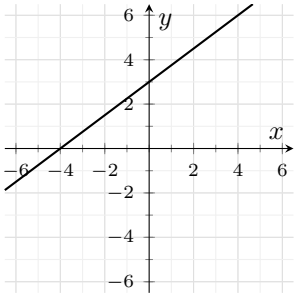
(d) $f(x) = -3x - 6$



(e) $f(x) = -4$

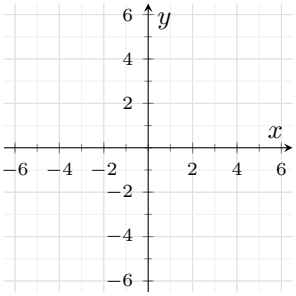


(f) $f(x) = \frac{3}{4}x + 3$

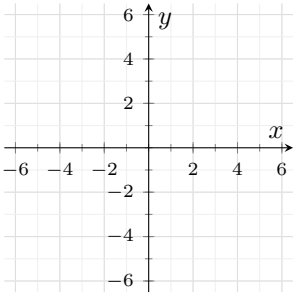


3. Sketch each graph on the grid without a table. Label the vertex and the y-intercept.

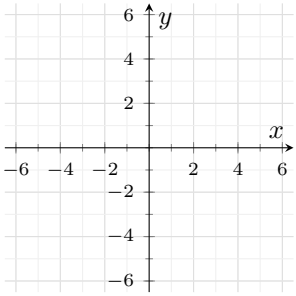
(a) $y = |2x - 5|$



(b) $y = |-x - 3|$



(c) $y = \left|\frac{1}{2}x + 2\right|$



4. For each function, state the vertex, the y-intercept, the domain and the range.

(a) $y = |4x + 12|$

(b) $y = |-5x + 10|$

(c) $y = \left|\frac{2}{3}x - 4\right|$

(d) $y = |7 - 0.5x|$

5. Let $f(x) = -2x - 8$. Complete the table to compare $y = f(x)$ with $y = |f(x)|$.

	$y = f(x)$	$y = \|f(x)\|$
Domain		
Range		
x-intercept		
y-intercept		

6. Write each function in piecewise form.

(a) $y = |x - 7|$

(b) $y = |3x + 9|$

(c) $y = |5 - x|$

(d) $y = |8 - 4x|$

(e) $y = |-2x - 5|$

(f) $y = \left|\frac{1}{3}x - 2\right|$

7. True or false? Explain.

(a) $|x+8| = x+8$ if $x \geq -8$

(b) $|4x-1| = 4x-1$ if $x < \frac{1}{4}$

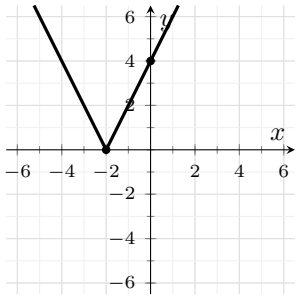
(c) $|7-2x| = 2x-7$ if $x > \frac{7}{2}$

(d) $|3x+2| = -3x-2$ if $x \geq -\frac{2}{3}$

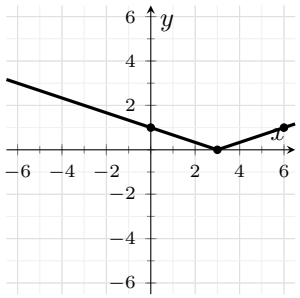
(e) $|10-5x| = 10-5x$ if $x \leq 2$

8. The graph of $y = |f(x)|$ is shown, where f is linear. Write an equation in the form $y = |ax+b|$, then write it in piecewise form.

(a)



(b)



9. Answer each question about a function f and the graph of $y = |f(x)|$.

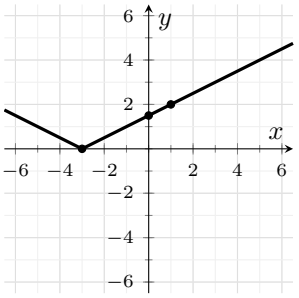
(a) The point $(5, -9)$ is on $y = f(x)$. State the matching point on $y = |f(x)|$.

(b) The point $(-1, 6)$ is on $y = f(x)$. State the matching point on $y = |f(x)|$.

(c) The point $(0, 4)$ is on $y = |g(x)|$. What are the possible values of $g(0)$?

(d) The x-intercept of $y = h(x)$ is -7. State the x-intercept of $y = |h(x)|$.

10. (a) The graph of $y = |f(x)|$, where f is linear, is shown. Sketch the two possible graphs of $y = f(x)$ on the grid and write their equations.



(b) A line $y = g(x)$ has slope -3, and the vertex of $y = |g(x)|$ is $(4, 0)$. Write the equation of g and state the y-intercept of $y = |g(x)|$.

11. Explain why the equation $|f(x)| = -2$ has no solution and why the inequality $|f(x)| < 0$ is never true. What is the range of $y = |f(x)|$ when $f(x) = -5$ for every x?

12. A drone flies at a constant 15 m/s along a straight line and passes directly over a camera. Its horizontal distance from the camera after t seconds is $d = |15t - 120|$ metres, for $0 \le t \le 14$.

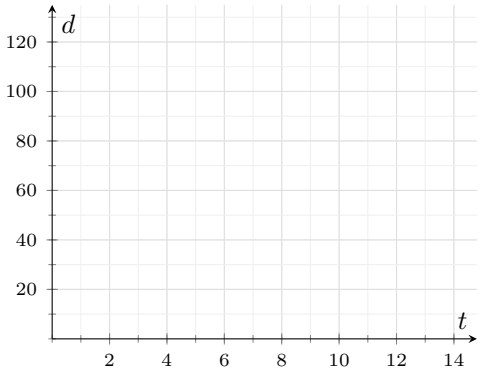
(a) Sketch the graph of d on the grid.

(b) When is the drone directly over the camera?

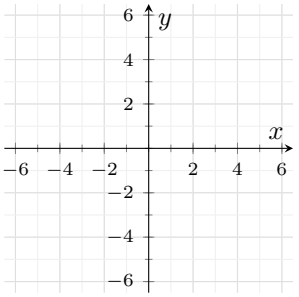
(c) How far from the camera was it at the start and at $t = 14$?

(d) Use the graph to find when it is 45 m from the camera.

(e) State the range of d in this context.



13. Sketch $y = |x - 2|$ and $y = |x| - 2$ on the same grid. Explain why the graphs are different, and state the range of each.



14. The graph of $y = f(x)$ passes through $(-2, -5)$ and $(6, 3)$. The graph of $y = |f(x)|$ must pass through

A. $(2, 5)$ and $(6, 3)$ **B.** $(-2, 5)$ and $(6, 3)$ **C.** $(-2, 5)$ and $(6, -3)$ **D.** $(2, -5)$ and $(-6, 3)$

15. Let $f(x) = 3x - 12$. Which statement is **false**?

A. The graphs of $y = f(x)$ and $y = |f(x)|$ both have x-intercept 4.

B. The y-intercept of $y = |f(x)|$ is 12.

C. The range of $y = |f(x)|$ is $\{y \mid y \geq 0, y \in \mathbb{R}\}$.

D. $|f(x)| = 3x - 12$ if $x < 4$.

16. The piecewise form of $y = |5 - 2x|$ is

A. $y = \begin{cases} 5 - 2x, & x \leq \frac{5}{2} \\ 2x - 5, & x > \frac{5}{2} \end{cases}$ **B.** $y = \begin{cases} 5 - 2x, & x \geq \frac{5}{2} \\ 2x - 5, & x < \frac{5}{2} \end{cases}$

C. $y = \begin{cases} 5 - 2x, & x \leq -\frac{5}{2} \\ 2x - 5, & x > -\frac{5}{2} \end{cases}$ **D.** $y = \begin{cases} 5 - 2x, & x \leq \frac{5}{2} \\ 2x + 5, & x > \frac{5}{2} \end{cases}$

17. Numerical Response The graph of $y = |-4x + 10|$ has its vertex at $(a, 0)$ and its y-intercept at $(0, b)$. The value of $a + b$, to the nearest tenth, is _____.

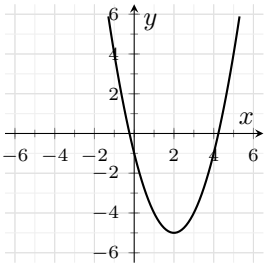
18. Numerical Response A line $y = f(x)$ has slope 2 and passes through $(1, -5)$. The graph of $y = |f(x)|$ touches the x-axis at $x =$ _____.

Lesson 3: Graphing Absolute Value Functions of Quadratic Functions

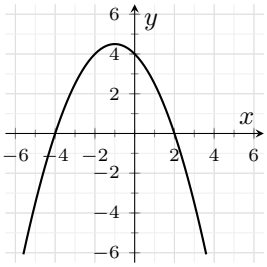
Assignment

1. The graph of $y = f(x)$ is shown. Sketch the graph of $y = |f(x)|$ on the same grid and state the invariant points, if any.

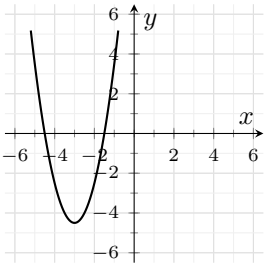
(a)



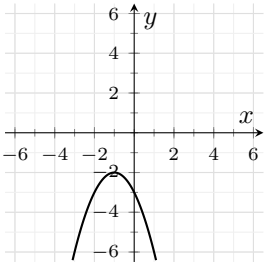
(b)



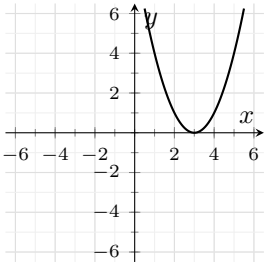
(c)



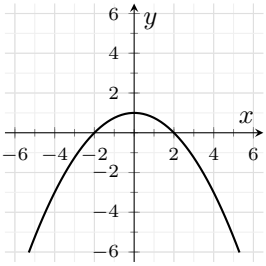
(d)



(e)



(f)



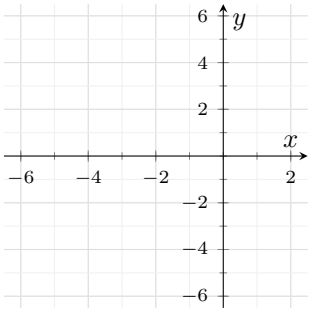
2. Consider $g(x) = -x^2 - 4x$.

(a) Complete the table of values.

x	-5	-4	-3	-2	-1	0	1
$g(x)$							
$\lvert g(x)\rvert$							

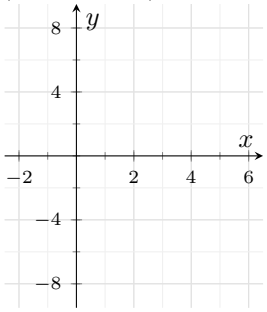
(b) Sketch $y = g(x)$ and $y = |g(x)|$ on the grid.

(c) State the domain and range of each function.

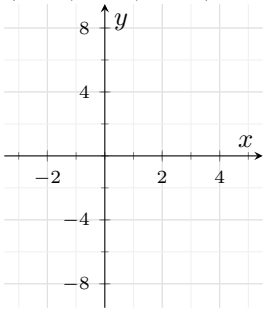


3. Sketch the graph of each function. Label the invariant points and the vertex of the graph.

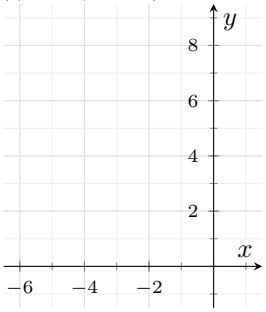
(a) $y = |x^2 - 4x - 5|$



(b) $y = |-2(x-1)^2 + 8|$



(c) $y = |(x+3)^2 + 2|$



4. For each function, state the invariant points (if any), the y-intercept, the vertex of the graph, and the range. Do not graph.

(a) $y = |x^2 - 10x + 21|$

(b) $y = |-\frac{1}{3}(x-6)^2 - 1|$

(c) $y = |3x^2 + 6x|$

(d) $y = |-(x+4)^2 + 1|$

5. $f(x) = 2x^2 - 4x - 6$. Compare $y = f(x)$ and $y = |f(x)|$.

	$y = f(x)$	$y = \|f(x)\|$
Domain		
Range		
x-intercepts		
y-intercept		
Vertex / turning point		

6. Write each function in piecewise form, without absolute value symbols.

(a) $y = |x^2 - 3x - 10|$

(b) $y = |25 - 4x^2|$

(c) $y = |x^2 + 8x + 16|$

(d) $y = | - x^2 + 4x - 7|$

7. Explain why $y = |x^2 + 2x + 5|$ can be written without absolute value symbols as a single piece.

8. True or false? If false, correct the statement.

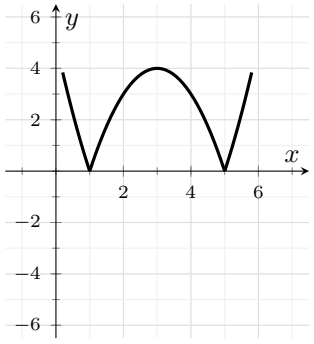
(a) $|x^2 - 49| = x^2 - 49$ if $x \geq 7$

(b) $|x^2 - 4x| = 4x - x^2$ if $x < 0$

(c) The graphs of $y = |x^2 - 2x - 8|$ and $y = |8 + 2x - x^2|$ are the same.

(d) If f is quadratic, the graph of $y = |f(x)|$ always has two invariant points.

9. The graph of $y = |f(x)|$ is shown.



(a) f is a quadratic function. Sketch the two possible graphs of $y = f(x)$ on the grid and write each equation in vertex form.

(b) Write $y = |f(x)|$ in piecewise form, using the first equation from (a).

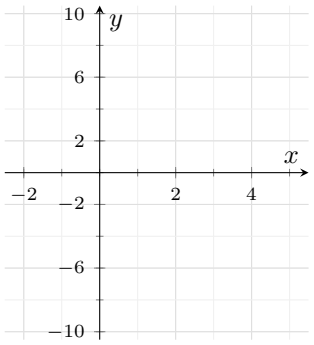
(c) Suppose f is any continuous function, not necessarily quadratic. Describe one more graph of $y = f(x)$ that gives this same graph of $y = |f(x)|$.

10. A quadratic function h has range $\{y \mid y \geq -2, y \in \mathbb{R}\}$, x-intercepts 1 and 3, and y-intercept 6.

(a) Sketch the graph of $y = |h(x)|$ on the grid.

(b) State the range of $y = |h(x)|$ and the coordinates of the vertex of its middle arc.

(c) Write an equation for h.



11. Consider $f(x) = (x+2)^2 + q$.

(a) For which values of q is the graph of $y = |f(x)|$ identical to the graph of $y = f(x)$?

(b) Determine q if the range of $y = |f(x)|$ is $\{y \mid y \geq 6, y \in \mathbb{R}\}$.

(c) State the range of $y = |f(x)|$ when $q = -5$, and the number of invariant points.

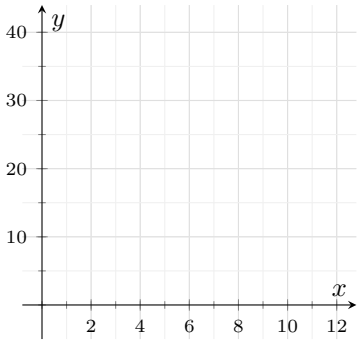
(d) $g(x) = -(x+2)^2 + k$ and the range of $y = |g(x)|$ is $\{y \mid y \geq 3, y \in \mathbb{R}\}$. Determine k.

12. A small business models its monthly profit by $P(x) = -x^2 + 14x - 40$, where P is in hundreds of dollars and x is the number of workshops it runs, $0 \leq x \leq 12$. A negative profit is a loss. The size of the profit or loss is $|P(x)|$.

(a) For which values of x is $|P(x)| = 0$?

(b) Sketch $y = |P(x)|$ for $0 \leq x \leq 12$.

(c) State the greatest profit and the greatest loss, and the range of $y = |P(x)|$ for $0 \leq x \leq 12$.



13. The graph of $y = f(x)$, a parabola, passes through $(-3, -10)$ and has x-intercepts -1 and 5. Which point must be on the graph of $y = |f(x)|$?

A. $(3, -10)$ **B.** $(-3, 10)$ **C.** $(-10, 3)$ **D.** $(3, 10)$

14. The range of $y = |-2(x-3)^2 - 4|$ is

A. $\{y \mid y \geq 0, y \in \mathbb{R}\}$ **B.** $\{y \mid y \leq -4, y \in \mathbb{R}\}$ **C.** $\{y \mid y \geq 4, y \in \mathbb{R}\}$ **D.** $\{y \mid y \geq -4, y \in \mathbb{R}\}$

15. Which statement is **false**?

A. If $y = f(x)$ has y-intercept -6, then $y = |f(x)|$ has y-intercept 6.

B. If a quadratic $y = f(x)$ has vertex $(2, -7)$ and opens up, then $y = |f(x)|$ has the point $(2, 7)$.

C. If a quadratic $y = f(x)$ has no x-intercepts, then $y = |f(x)|$ has the same graph as $y = f(x)$.

D. The domain of $y = |f(x)|$ is the same as the domain of $y = f(x)$.

16. Numerical Response The graph of $y = |x^2 - 2x - 15|$ has a middle arc between its invariant points. The y-coordinate of the vertex of this arc is _____.

17. Numerical Response The horizontal line $y = k$ meets the graph of $y = |x^2 - 8x + 12|$ at exactly three points. The value of k is _____.

Lesson 4: Solving Absolute Value Equations (Linear)

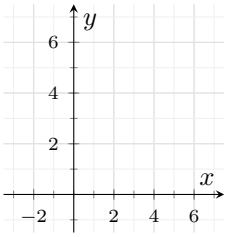
Assignment

1. Solve.
(a) $|x| = 12$ (b) $|x + 6| = 2$ (c) $|x - 9| = 0$ (d) $|x| = -4$

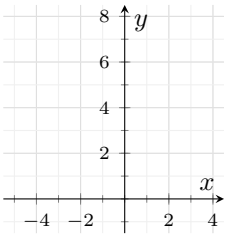
2. Describe how to use the intersection method on a graphing calculator to solve $|2x - 3| = x + 3$. State the solution.

3. Describe how to use the zero method to solve $|x + 1| = 2x - 4$. State the solution.

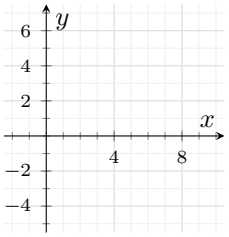
4. Solve each equation graphically. Sketch and label both graphs.
(a) $|x - 2| = 4$



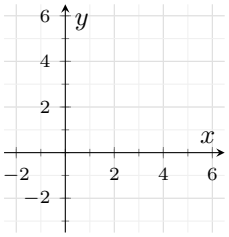
(b) $|2x + 2| = x + 4$



(c) $|x - 3| = \frac{1}{2}x - 4$



(d) $|3 - x| = x - 1$



5. Solve algebraically.
(a) $|x - 7| = 10$ (b) $|5x + 2| = 13$ (c) $|6 - 2x| = 0$

(d) $|3x - 4| + 9 = 5$ (e) $4|x + 1| - 3 = 17$ (f) $-3|2x - 1| = -15$

(g) $\left|\frac{1}{2}x - 3\right| = 1$ (h) $7 - |x + 2| = 3$ (i) $\frac{|4x-1|}{3} = 5$

6. Jordan solved $|x-3|-5=2$ like this: $x-3-5=2$ or $x-3-5=-2$, so $x=10$ or $x=6$. Find the error, show that one of Jordan's answers is wrong, and solve the equation correctly.

7. Solve by cases. Show the number line, both cases and whether each answer is in its subdomain.

(a) $|x+2|=3x$

(b) $|2x-6|=x$

(c) $|5-x|=2x+1$

(d) $|3x+1|=x-3$

8. Solve and verify. Identify any extraneous solutions.

(a) $3|x-2|=x+6$

(b) $|4x-3|-2x=5$

(c) $|x-6|+2x=0$

(d) $|2x+5|=3x+10$

9. Solve $\left|\frac{1}{2}x+1\right|=x-2$.

10. A home thermostat is set to 20 °C. The furnace turns on or off whenever the room temperature differs from the setting by 1.5 °C. Write and solve an absolute value equation for these two temperatures.

11. Write an absolute value equation, in the form $|x - a| = d$, with the given solutions.
(a) -2 and 10 (b) -7.5 and -1.5 (c) 5 only

12. The distance between a number and 7 is three times the number. Write an equation and determine the number.

13. Consider $|x - 3| = k$ and $|2x + 1| + k = 4$.
(a) For which values of k does $|x - 3| = k$ have two solutions? one solution? no solution?

(b) For which values of k does $|2x + 1| + k = 4$ have no solution?

14. The complete solution of $2|x - 1| + 3 = 11$ is
A. $x = 5$ only **B.** $x = 5$ or $x = -3$ **C.** $x = 5$ or $x = -6$ **D.** $x = 3$ or $x = -5$

15. How many solutions does $|2x - 1| = x - 4$ have?
A. 0 **B.** 1 **C.** 2 **D.** infinitely many

16. Which equation has **no** solution?
A. $|x + 3| = 0$ **B.** $|x| + 2 = 0$ **C.** $|x - 1| - 4 = 0$ **D.** $-|x| = -5$

17. Numerical Response The sum of the solutions of $|3x - 12| = 2x + 2$ is _____.

18. Numerical Response The solution of $|5 - 2x| = 3x + 1$, to the nearest tenth, is _____.

Lesson 5: Solving Absolute Value Equations (Quadratic)

Assignment

1. Solve algebraically.

(a) $|x^2 - 7| = 9$

(b) $|x^2 - 3x| = 4$

(c) $|x^2 + 5x| = 6$

(d) $|x^2 - 8x + 12| = 0$

(e) $|x^2 + 9| = 5$

(f) $|x^2 - 4x| = -3$

2. Isolate the absolute value, then solve.

(a) $3|x^2 - 2x| + 1 = 10$

(b) $5 - |x^2 - 1| = 2$

(c) $\frac{1}{2}|x^2 + 2x - 1| = 1$

3. Solve. Give exact answers, then answers to the nearest hundredth.

(a) $|x^2 - 2x - 4| = 2$

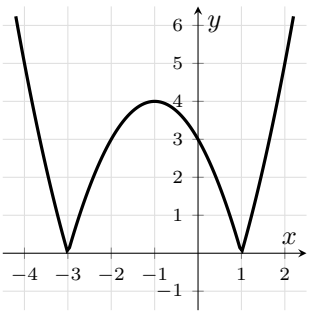
(b) $|2x^2 - x - 4| = 2$

(c) $|x^2 + 6x + 4| = 1$

4. The graph of $y = |x^2 + 2x - 3|$ is shown.

(a) Use the graph to solve $|x^2 + 2x - 3| = 3$: give the two exact solutions you can read, and estimate the other two.

(b) Solve the equation algebraically to confirm.



5. The parabola $y = x^2 - 4x - 5$ has zeros -1 and 5 and vertex $(2, -9)$. State the number of solutions of $|x^2 - 4x - 5| = k$ for each value of k, and explain using a sketch of $y = |x^2 - 4x - 5|$.
(a) $k = -1$ (b) $k = 0$ (c) $k = 4$ (d) $k = 9$ (e) $k = 12$

6. For what value of k does $|x^2 + 2x - 8| = k$ have exactly three solutions? Solve the equation for that value of k.

7. Solve and check. Reject any extraneous solutions.
(a) $|x^2 - 6| = x$

(b) $|x^2 - 2x| = 2x - 3$

(c) $|x - 4| = x^2 - 6x + 8$

(d) $|2x + 3| = x + 1$

8. A student solved $|x^2 - 5| = x + 1$ and wrote: "$x = 3,\ -2,\ \frac{-1+\sqrt{17}}{2},\ \frac{-1-\sqrt{17}}{2}$." Identify the error and give the correct solution.

9. Solve.
(a) $|x - 4| = |3x + 2|$

(b) $|x^2 - 2x| = |x + 4|$

(c) $|x^2 - 4| = |x^2 - 2x|$

10. Solve using subdomains.

(a) $|x+2|+|x-4|=10$

(b) $|2x-1|-|x+1|=1$

11. Use a graphing calculator to solve $|0.5x^2-2x-3|=4$ to the nearest tenth. Describe the window and the method you used, then confirm one solution with the quadratic formula.

12. The temperature in a greenhouse t hours after 6:00 a.m. is $T=-0.5t^2+6t+10$ degrees Celsius, $0\le t\le 12$. The ideal temperature is 24 °C.

(a) Write an absolute value equation for the times when the temperature is exactly 2 °C away from ideal, and solve it exactly.

(b) State the clock times to the nearest minute. Which of these times have a temperature of 26 °C?

13. How many solutions does $|x^2+4x-5|=9$ have?

A. 2 **B.** 3 **C.** 4 **D.** 0

14. The solution of $|x^2-x|=x-1$ is

A. $x=1$ only **B.** $x=-1$ or $x=1$ **C.** $x=0$ or $x=1$ **D.** no real solution

15. Which equation has no real solution?

A. $|x^2-4|=0$ **B.** $|x^2+1|=1$ **C.** $|x^2+3|=2$ **D.** $|x^2-9|=5$

16. Numerical Response The largest solution of $|x^2 + 3x - 2| = 5$, to the nearest hundredth, is _____.

17. Numerical Response The largest solution of $|x^2 - 3| = |x^2 - 4x + 1|$, to the nearest hundredth, is _____.

Lesson 6: Reciprocal Functions of Linear Functions

Assignment

1. Complete the table of reciprocals.

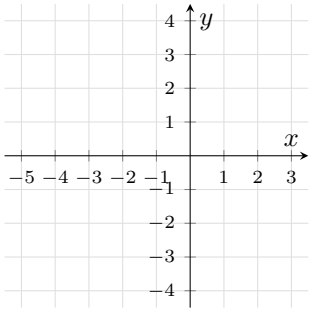
$f(x)$	5	-2	$\frac{1}{4}$	-0.1	1	-1	0	100
$\frac{1}{f(x)}$								

2. Let $f(x) = x + 1$.

(a) Complete the table.

x	-5	-3	-2	-1.5	-1	-0.5	0	1	3
$f(x)$									
$\frac{1}{f(x)}$									

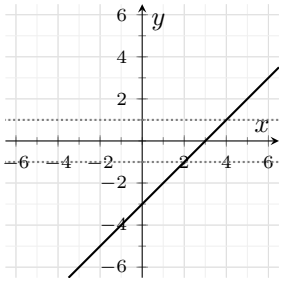
(b) Sketch $y = f(x)$ and $y = \dfrac{1}{f(x)}$ on the grid.



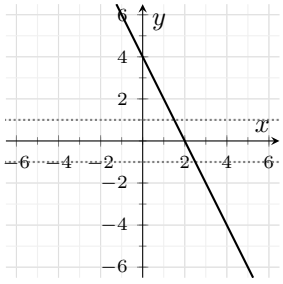
(c) State the equations of the asymptotes, the invariant points, and the y-intercept of $y = \dfrac{1}{f(x)}$.

3. Sketch the graph of $y = \dfrac{1}{f(x)}$ on each grid. State the vertical asymptote and the y-intercept of the reciprocal graph.

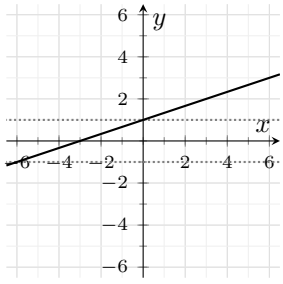
(a) $f(x) = x - 3$



(b) $f(x) = -2x + 4$



(c) $f(x) = \frac{1}{3}x + 1$



4. Complete the table without graphing.

	(a) $y = \dfrac{1}{x-5}$	(b) $y = \dfrac{1}{4x+2}$	(c) $y = \dfrac{1}{6-3x}$	(d) $y = \dfrac{1}{\frac{1}{2}x-2}$
Vertical asymptote				
y-intercept				
Invariant points				
Domain				
Range				

5. Consider $y = \dfrac{1}{x+1}$.

(a) Describe what happens to y as x approaches -1 from the right ($x = -0.9, -0.99, \dots$) and from the left ($x = -1.1, -1.01, \dots$).

(b) Describe y as x becomes very large and positive, and very large and negative. Why does the graph never touch the x-axis?

6. Answer each question.

(a) The graph of $y = f(x)$ passes through $(3, 8)$. State the corresponding point on $y = \dfrac{1}{f(x)}$.

(b) The graph of $y = \dfrac{1}{f(x)}$ passes through $(-1, -0.4)$. State the corresponding point on $y = f(x)$.

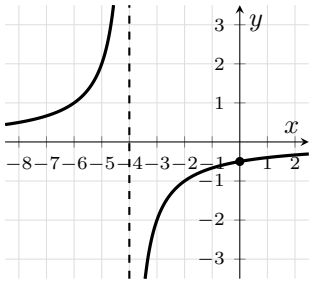
(c) For $f(x) = 3x - 1$, determine the points on $y = \dfrac{1}{f(x)}$ where $y = 2$ and where $y = -\frac{1}{2}$.

(d) For $f(x) = 4 - 2x$, for which values of x is $\dfrac{1}{f(x)}$ positive? Does each branch of $y = \dfrac{1}{f(x)}$ rise or fall from left to right? Explain.

7. The graph of $y = \dfrac{1}{f(x)}$, where f is linear, is shown. It has the vertical asymptote $x = -4$ and the y-intercept $-\frac{1}{2}$.

(a) Sketch $y = f(x)$ on the grid.

(b) Determine $f(x)$ in the form $f(x) = ax + b$.



8. The graph of $y = \dfrac{1}{f(x)}$, where f is linear, has the vertical asymptote $x = 3$ and passes through $(5, 0.25)$. Determine the equation of $f(x)$.

9. A student says: "The graph of $y = \dfrac{1}{2x + 6}$ has the vertical asymptote $x = -6$ and passes through the invariant point $(-2.5, 1)$." Which part is correct? Correct the error.

10. Explain.

(a) Why does the reciprocal of a non-horizontal linear function always have exactly one vertical asymptote?

(b) Describe the graph of $y = \dfrac{1}{f(x)}$ when $f(x) = 2$ (a horizontal line). Does it have a vertical asymptote or invariant points?

11. (a) The graph of $y = \dfrac{1}{kx + 1}$ passes through $(2, -1)$. Determine k and the vertical asymptote.

(b) The invariant points of $y = \dfrac{1}{f(x)}$, f linear, are $(1, 1)$ and $(3, -1)$. Determine $f(x)$ and the vertical asymptote.

12. The vertical asymptote of the graph of $y = \dfrac{1}{3x + 12}$ is

A. $x = 12$ **B.** $x = -4$ **C.** $x = 4$ **D.** $y = -4$

13. Which statement about the graph of $y = \dfrac{1}{2x - 1}$ is **false**?

A. The vertical asymptote is $x = \frac{1}{2}$.
B. $(0, -1)$ is an invariant point.
C. $y > 0$ when $x > \frac{1}{2}$.
D. For $x > \frac{1}{2}$, y increases as x increases.

14. The graph of $y = \dfrac{1}{f(x)}$, f linear, has the vertical asymptote $x = -3$ and passes through $(-2, 1)$. The equation of f is

A. $f(x) = x - 3$ **B.** $f(x) = x + 3$ **C.** $f(x) = -x - 3$ **D.** $f(x) = 3x + 9$

15. Numerical Response For $f(x) = \frac{1}{4}x + 1$, the graph of $y = \dfrac{1}{f(x)}$ passes through $(6, p)$. The value of p, to the nearest tenth, is _____.

16. Numerical Response The graph of $y = \dfrac{1}{4x - 10}$ has invariant points $(a, 1)$ and $(b, -1)$. The value of $a + b$ is _____.

Lesson 7: Reciprocal Functions of Quadratic Functions

Assignment

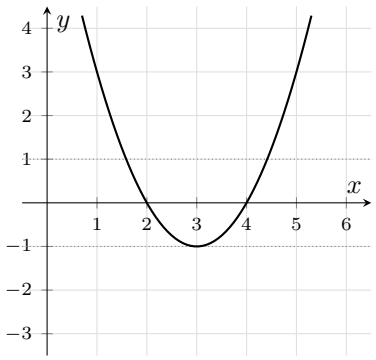
1. Let $f(x) = x^2 - 6x + 8$. Its graph is shown.

(a) Complete the table, then sketch $y = \dfrac{1}{f(x)}$ on the grid.

(b) State the equations of the asymptotes and the coordinates of the turning point of $y = \dfrac{1}{f(x)}$.

(c) State the invariant points (exact values).

x	0	1	2	3	4	5	6
$f(x)$							
$\dfrac{1}{f(x)}$							



2. State the number of vertical asymptotes of the graph of $y = \dfrac{1}{f(x)}$ and their equations.

(a) $f(x) = x^2 + x - 12$

(b) $f(x) = 4x^2 - 12x + 9$

(c) $f(x) = x^2 - 2x + 7$

(d) $f(x) = -3x^2 + 5x + 2$

(e) $f(x) = x^2 - 5$

(f) $f(x) = 2(x + 3)^2 + 1$

3. For each function, state the turning point of $y = \dfrac{1}{f(x)}$, whether it is a maximum or a minimum, and the domain and range of $y = \dfrac{1}{f(x)}$.

(a) $f(x) = (x - 1)^2 + 5$

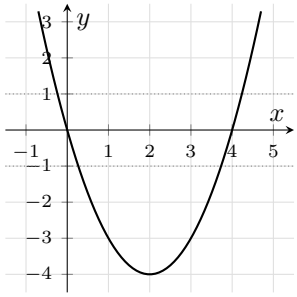
(b) $f(x) = -(x + 2)^2 + 2$

(c) $f(x) = 2(x - 4)^2 - \frac{1}{2}$

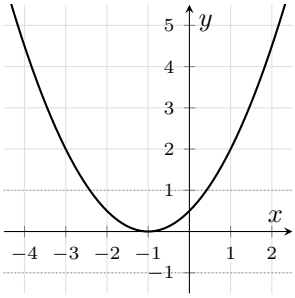
(d) $f(x) = -\frac{1}{2}x^2 - 4$

4. The graph of $y = f(x)$ is shown. Sketch the graph of $y = \dfrac{1}{f(x)}$ on the same grid, with its asymptotes, and label the turning point (if any).

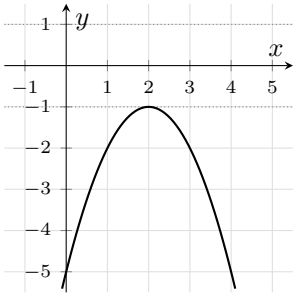
(a) $f(x) = x^2 - 4x$



(b) $f(x) = \frac{1}{2}(x+1)^2$



(c) $f(x) = -(x-2)^2 - 1$



5. Let $f(x) = x^2 + 2x - 8$. For $y = \dfrac{1}{f(x)}$, determine:

(a) the equations of the asymptotes

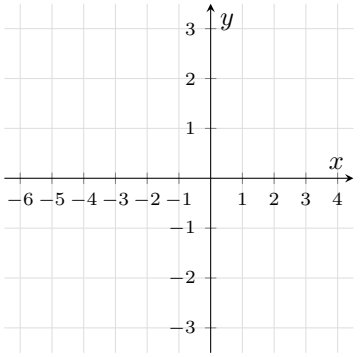
(b) the y-intercept

(c) the turning point and its type

(d) the invariant points (exact)

(e) the domain and range

Then sketch $y = \dfrac{1}{f(x)}$ on the grid.

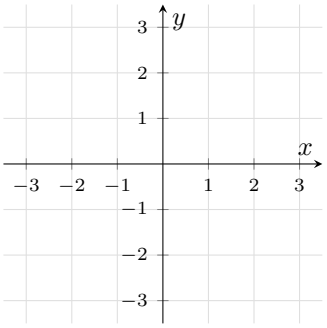


6. Let $f(x) = x^2 - 3$.

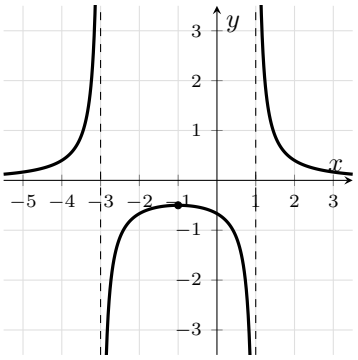
(a) Write the equation of the reciprocal function.

(b) Use a graphing calculator to sketch $y = f(x)$ and its reciprocal on the grid.

(c) Determine the exact coordinates of the four invariant points.



7. The graph of $y = g(x)$, where $g(x) = \dfrac{1}{f(x)}$ and f is a quadratic function, is shown. It has vertical asymptotes $x = -3$ and $x = 1$ and a maximum point $\left(-1, -\frac{1}{2}\right)$.



(a) Describe a strategy for sketching $y = f(x)$, then sketch it on the grid.

(b) Write $f(x)$ in vertex form and in general form.

(c) State the y-intercept of $y = g(x)$.

8. The graph of $y = \dfrac{1}{f(x)}$, where f is a quadratic function, has a maximum point $\left(3, \frac{1}{2}\right)$ and a y-intercept of $\frac{1}{20}$. Determine $f(x)$ in vertex form and in general form.

9. (a) The point $(5, 4)$ is on $y = f(x)$. State the matching point on $y = \dfrac{1}{f(x)}$.

(b) The point $(-2, -0.125)$ is on $y = \dfrac{1}{f(x)}$. State the matching point on $y = f(x)$.

(c) The point $(6, 1)$ is on $y = f(x)$. What can you say about it?

(d) Let $f(x) = x^2 - 2x - 6$. The point $\left(a, \frac{1}{9}\right)$, with $a > 0$, is on $y = \dfrac{1}{f(x)}$. Determine a.

10. Let $f(x) = x^2 + 4x + 7$.

(a) Explain why the graph of $y = \dfrac{1}{f(x)}$ has no vertical asymptotes.

(b) Explain why the graph of $y = \dfrac{1}{f(x)}$ has no points in quadrants 3 and 4.

(c) State the maximum value and the range of $y = \dfrac{1}{f(x)}$. Does the graph have any invariant points?

11. Let $f(x) = (x + 1)^2 - 4$.

(a) State the intervals on which f is increasing and decreasing.

(b) State the intervals on which $y = \dfrac{1}{f(x)}$ is increasing and decreasing. (Remember the asymptotes.)

12. (a) Liam says: "A quadratic has two zeros, so the graph of $y = \dfrac{1}{x^2 - 6x + 9}$ has two vertical asymptotes." Explain his error.

(b) Maya says: "The vertex of $f(x) = x^2 + 6x + 10$ is $(-3, 1)$, so the graph of $y = \dfrac{1}{f(x)}$ has a minimum point at $(-3, 1)$." Explain her error.

13. A student makes these statements about $y = \dfrac{1}{f(x)}$, where f is a quadratic function.

Statement 1: If f has exactly one x-intercept, the reciprocal graph has exactly one vertical asymptote.
Statement 2: If $f(x) = (x - 2)^2 - 5$, the reciprocal graph has a minimum point at $\left(2, -\frac{1}{5}\right)$.
Statement 3: The reciprocal graph never meets the x-axis.
The true statements are

A. 1 and 2 only **B.** 1 and 3 only **C.** 2 and 3 only **D.** 1, 2 and 3

14. The graph of $y = \dfrac{1}{f(x)}$ has vertical asymptotes $x = -2$ and $x = 3$ and a y-intercept of $-\frac{1}{6}$. The function f could be

A. $x^2 - x - 6$ **B.** $x^2 + x - 6$ **C.** $-x^2 + x + 6$ **D.** $2x^2 - 2x - 12$

15. The range of the quadratic function $y = f(x)$ is $\{y \mid y \leq 4, y \in \mathbb{R}\}$. The range of $y = \dfrac{1}{f(x)}$ is

A. $\left\{y \mid y < 0 \text{ or } y \geq \frac{1}{4}, y \in \mathbb{R}\right\}$
B. $\left\{y \mid 0 < y \leq \frac{1}{4}, y \in \mathbb{R}\right\}$
C. $\left\{y \mid y \leq \frac{1}{4}, y \neq 0, y \in \mathbb{R}\right\}$
D. $\left\{y \mid y \geq \frac{1}{4}, y \in \mathbb{R}\right\}$

16. The number of invariant points of the graphs of $f(x) = x^2 - 4x + 5$ and $y = \dfrac{1}{f(x)}$ is

A. 0 **B.** 1 **C.** 2 **D.** 4

17. Numerical Response The graph of $y = \dfrac{1}{x^2 + kx + 9}$, where $k > 0$, has exactly one vertical asymptote. The value of k is _____.

18. Numerical Response The graph of $y = \dfrac{1}{f(x)}$, where f is a quadratic function, has a maximum point $(1, 0.25)$ and a y-intercept of 0.2. The value of $f(3)$ is _____.

Lesson 8: Unit 7 Review

Assignment

1. Evaluate.
(a) $|-12| + |4 - 9|$ (b) $-|-3|^2$ (c) $|\sqrt[3]{-8} - 5|$ (d) $\big|\,|-7| - |10|\,\big|$

2. (a) Order from least to greatest: $|-2.4|,\ -|-3|,\ |1-5|,\ |\sqrt{5}-3|,\ -|0.5|$

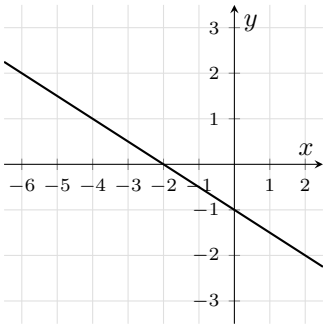
(b) Determine the distance between -3.2 and -11.7 on a number line.

(c) Two numbers are each 6 units from 2 on a number line. Write an absolute value equation and determine the numbers.

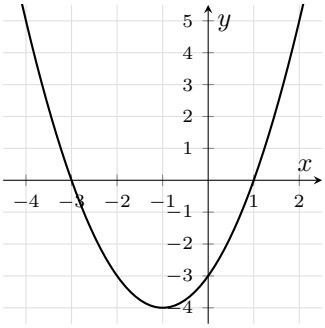
3. Write in piecewise form.
(a) $y = |x - 6|$ (b) $y = |3 - 4x|$ (c) $y = |x^2 - 9|$

4. The graph of $y = f(x)$ is shown. Sketch $y = |f(x)|$ on the same grid. State the invariant points, the y-intercept and the range of $y = |f(x)|$.

(a) $f(x) = -\frac{1}{2}x - 1$



(b) $f(x) = x^2 + 2x - 3$



5. For $y = |x^2 - 6x + 5|$, state the x-intercepts, the y-intercept, the coordinates of the local maximum point, the domain and the range.

6. Solve algebraically and verify.
(a) $|x-3|=7$ (b) $2|x+1|-5=7$ (c) $|4x+1|+9=2$

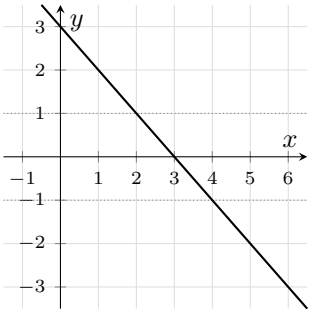
(d) $|2x-1|=x+4$ (e) $|x+5|=2x+1$

7. Solve algebraically. Identify any extraneous solutions.
(a) $|x^2-12|=4$ (b) $|x^2-x-6|=x+2$ (c) $|x^2-2x|=3$

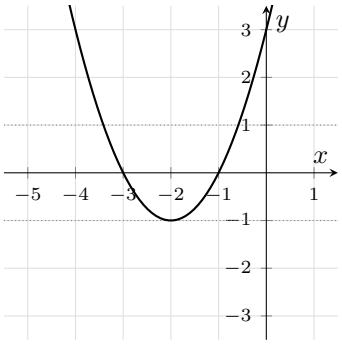
8. Use a graphing calculator to solve $|x^2-4x|=2$ to the nearest tenth. Describe your method. Then verify one solution exactly with the quadratic formula.

9. The graph of $f(x)=3-x$ is shown.
(a) Write the equation of the reciprocal function.

(b) Sketch the reciprocal. State its asymptotes, invariant points and y-intercept, and its domain and range.



10. The graph of $f(x) = x^2 + 4x + 3$ is shown. Sketch $y = \dfrac{1}{f(x)}$ on the grid. State the asymptotes, the turning point and its type, the y-intercept, the exact invariant points, and the domain and range.



11. The graph of $y = \dfrac{1}{f(x)}$, where f is a quadratic function, has a maximum point $(-1, 0.2)$ and a y-intercept of $\frac{1}{7}$. Determine $f(x)$ in general form.

12. Let $f(x) = x^2 - 9$. Complete the table.

	$y = f(x)$	$y = \lvert f(x)\rvert$	$y = \dfrac{1}{f(x)}$
Domain			
Range			

13. Explain why the graph of $y = \dfrac{1}{f(x)}$ never crosses the x-axis, and why the equation $|f(x)| = -3$ has no solution.

14. Which expression has the greatest value?

A. $|-7| - |-2|$ **B.** $|-7-2|$ **C.** $|7| - 2|-2|$ **D.** $-|-9|$

15. The piecewise form of $y = |2x + 8|$ is

A. $y = \begin{cases} 2x+8, & x \geq -4 \\ -2x-8, & x < -4 \end{cases}$ **B.** $y = \begin{cases} 2x+8, & x \geq 4 \\ -2x+8, & x < 4 \end{cases}$ **C.** $y = \begin{cases} 2x+8, & x \geq -4 \\ 2x-8, & x < -4 \end{cases}$ **D.** $y = \begin{cases} 2x+8, & x \leq -4 \\ -2x-8, & x > -4 \end{cases}$

16. The point $(4, -0.5)$ is on the graph of $y = f(x)$. Which point must be on the graph of $y = \left|\dfrac{1}{f(x)}\right|$?

A. $(4, 2)$ **B.** $(4, -2)$ **C.** $(-4, 2)$ **D.** $(4, 0.5)$

17. The equation $|x + 1| = k$ has exactly one solution when

A. $k < 0$ **B.** $k = 0$ **C.** $k > 0$ **D.** $k = 1$

18. For which function is the graph of $y = |f(x)|$ identical to the graph of $y = f(x)$?

A. $x^2 - 1$ **B.** $(x-1)^2$ **C.** $-x^2$ **D.** $2x + 1$

19. The range of the quadratic function $y = f(x)$ is $\{y \mid y \geq 2, y \in \mathbb{R}\}$. The range of $y = \dfrac{1}{f(x)}$ is

A. $0 < y \leq \frac{1}{2}$ **B.** $y \geq \frac{1}{2}$ **C.** $y \leq \frac{1}{2}, y \neq 0$ **D.** $y < 0$ or $y \geq \frac{1}{2}$

20. The graph of $y = \dfrac{1}{x^2 - 2x + 1}$ has

A. two vertical asymptotes, $x = 1$ and $x = -1$

B. one vertical asymptote, $x = 1$

C. one vertical asymptote, $x = -1$

D. no vertical asymptotes

21. A function f is negative and increasing on the interval $1 < x < 4$. On this interval,

A. $y = |f(x)|$ and $y = \dfrac{1}{f(x)}$ are both decreasing

B. $y = |f(x)|$ is increasing and $y = \dfrac{1}{f(x)}$ is decreasing

C. $y = |f(x)|$ is decreasing and $y = \dfrac{1}{f(x)}$ is increasing

D. $y = |f(x)|$ and $y = \dfrac{1}{f(x)}$ are both increasing

22. The number of solutions of the equation $|x^2 - 2x - 6| = 2$ is

A. 1 **B.** 2 **C.** 3 **D.** 4

23. Numerical Response The value of $|-9| - |4 - 11| + |\sqrt[3]{-125}|$ is _____.

24. Numerical Response The sum of all the solutions of $|x^2 - 5x| = 6$ is _____.

25. Numerical Response The equation $|x - 5| = 2x + 1$ has one solution. To the nearest hundredth, it is _____.

26. Numerical Response The graph of $y = \dfrac{1}{f(x)}$, where f is quadratic, has vertical asymptotes $x = 0$ and $x = 4$ and a maximum point $(2, -0.5)$. To the nearest tenth, the y-coordinate of the point on this graph where $x = 5$ is _____.
